Dhruv Kapri

Applied AI/ML | AEC Technology | LLM Applications | Design Automation
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Education

Indian Institute of Technology, Kharagpur — West Bengal, India
Bachelor of Architecture: 8.31/10 — Dec 2021 – Apr 2026

Experience

SLnko Energy Pvt. Ltd., India | *Graduate Trainee - Product Engineer* — May 2026 – Present

- Building a **floorplan automation engine**: **CP-SAT** lays out rooms & the column grid as NBC/bylaw-checked plans
- **Site intake**: **PostGIS** point-in-polygon zoning and seismic lookup applies **20** bylaw rules to buildable plot polygons
- Built a **Tender Intelligence RAG app** (Docling + hybrid retrieval) cutting solar-EPC review from days to hours
- Building a deterministic **IS 456/1893 RCC structural engine** (PyNite FE) behind a licensed-engineer sign-off
- Co-authored **org-wide Revit BIM standards** adopted across architecture, structural & MEP on active projects

Veritas Group, Indonesia | *Digital Solutions Intern* — Feb 2025 – Jul 2025

- Prototyped **LLM-assisted tools** for summarization & design documentation using **Python** orchestration pipelines
- Built internal analytics & ML-assisted dashboards delivering live project insights, cutting stakeholder queries by **30%**
- Shipped Python-based parametric & automation tools for Rhino + Grasshopper, cutting design-iteration time by **40%**

OCAD University, Canada | *Mitacs Globalink Research Intern* — May 2024 – Aug 2024

- Built a persona-driven simulation framework using **112 LLM-based personas** to simulate urban planning scenarios
- Developed geospatial data pipelines to validate **61 amenities** across **3 districts**, enabling robust spatial modeling
- Applied travel-pattern analytics & unsupervised clustering on **158 case studies** to identify urban layout insights

Universiti Malaya, Malaysia | *Research Fellow* — Dec 2022 – Jan 2024

- Developed a **Vision Transformer** with contrastive learning: **98.58%** accuracy, **+5.8%** over a CNN baseline
- Built the data-augmentation & contrastive pre-training pipeline, improving feature separability & generalization
- Collected & analyzed a **12K+ image** dataset across positive, neutral & negative sentiments with **95% agreement**

Publications

- *Integrating AI-Driven Personas and Procedural Visualization for Complete Communities Design and Urban Planning for Large-Scale Urban Development*, **HCII 2025, Springer Nature Switzerland**
- *A New Contrastive Learning-based Vision Transformer for Sentiment Analysis using Scene Text Images*, **International Journal of Pattern Recognition and Artificial Intelligence (IJPRAI 2024)**

Projects

GIS-Driven Rooftop Solar Potential (SIH × ISRO) — *Live* | *GitHub*

- Built a **Rhino-Grasshopper + Ladybug** prototype: **LOD-1** model, full-year **roof & facade** radiation & shading
- Rebuilt as a city-scale **Python** pipeline: **LiDAR DSM**, **GRASS r.sun**, **RANSAC**, **PVWatts** yield; ∼1% off **Esri**
- Shipped a live **MapLibre + PMTiles** web map scoring **100K+** DC rooftops into a census-tract **equity** choropleth

End-to-End ML Pipeline for Vehicle Insurance — *GitHub*

- Built an end-to-end ML pipeline covering data ingestion, validation, feature engineering, training & evaluation
- Implemented automated CI/CD using **GitHub Actions** with deployment to **AWS (S3, EC2, ECR)** for production
- Served model inference through a **FastAPI** REST service using modular, reusable pipeline components throughout

Technical Skills

Languages: Python | C++ | JavaScript | TypeScript | SQL
ML & DL: PyTorch | TensorFlow | scikit-learn | Pandas | NumPy | Transformers | ViT | OpenCV
LLM & RAG: LangChain | LangGraph | LlamaIndex | RAG | Embeddings | Vector DBs | Docling
MLOps & Cloud: Git | Docker | GitHub Actions | AWS | FastAPI | CI/CD | Kubernetes | Ollama
AEC & Comp. Design: Revit | Dynamo | Rhino | Grasshopper | ArcGIS Pro | PostGIS | Ladybug

Positions of Responsibility

Design Captain, IIT Kharagpur | *Inter IIT Tech Meet 13.0* — Oct 2024 – Dec 2024

- Led a **20-member** design team, owning delivery across **12 problem-statement teams**, including the frontend UI
- Aligned **technical approach & research** across core teams with problem owners, mentors & contingent captains
- Shipped **30+ decks & 20+ technical reports** across evaluation milestones, enforcing accuracy & consistency